Unified Foundational Model in Waste Management

Evaluating Florence-2 Across Classification, Detection, and Segmentation Tasks

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Presentation Outline

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- Florence 2 : Workflow Pipeline
- Experiment Matrix (Models × Tasks × Datasets)
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Recap: Capstone Stage 1

Working on AI for Waste Management.

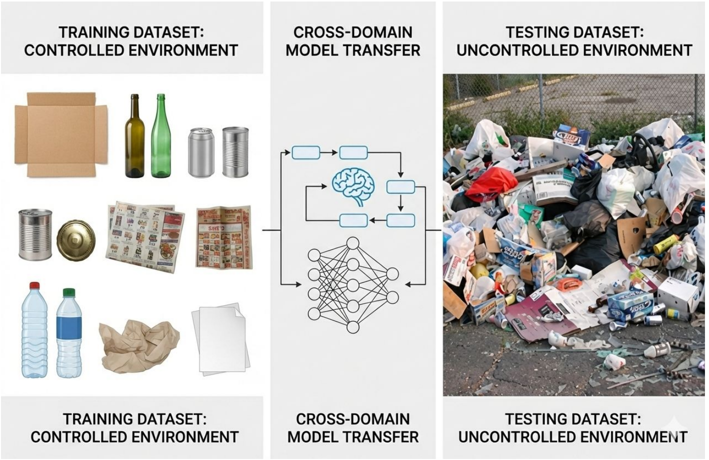
Problem Statement :

- Existing models are trained on clean and curated datasets.
- Real-world waste scenes differ significantly from training data.

  Domain shift arises due to:

  - Lighting variations
  - Cluttered backgrounds
  - Occlusions
  - Environmental changes

High lab accuracy does not guarantee real-world reliability.



Recap: Capstone Stage 1

Foundation Models vs Task-Specific Models

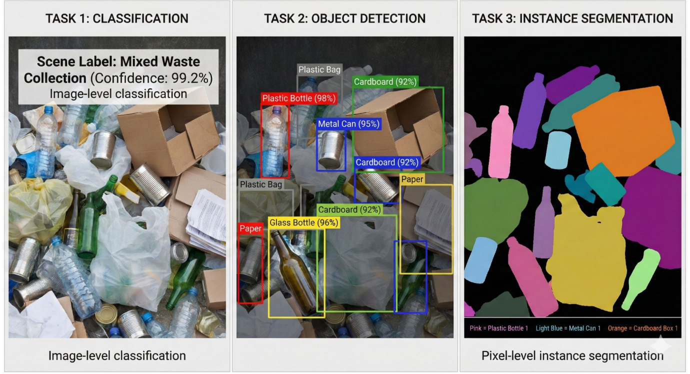
- Task-Specific Models **:** Task-Specific Models are trained for a single task with high accuracy in controlled Environment.
- Foundation Models **:** Foundation Models are trained on diverse large-scale data and support multiple tasks.

Foundation Models offer better robustness and cross-domain generalization

Tasks Considered: Classification, Detection & Segmentation

- Waste Classification → Identifies waste categories (plastic, paper)
- Waste Detection → Detects and localizes waste objects
- Waste Segmentation → Performs pixel-level waste separation

Evaluating all three tasks enables comprehensive real-world analysis



[Generalization abilities of foundation models in waste classification]

Recap: Capstone Stage 1

Controlled Environment Datasets

- **TrashNet** – Lab-based waste classification
- **TACO** – Urban waste detection
- **TACO Masks** – Real-world segmentation masks

Cross-Domain (Real-World) Datasets

- **RealWaste** – Real-world cluttered waste images
- **Trash-ICRA19** – Underwater waste scenarios
- **Dense Waste Segmentation Dataset** – Controlled segmentation scenes

Stage 1 Results:

Task	**In-Domain (Task-Specific)**	**Cross-Domain (Foundation)**
Classification	ViT-Base — 96.4%	CLIP — 42.7%
Detection	YOLOv8 — 64.2% mAP	Grounding DINO — 42.7% mAP
Segmentation	DeepLabV3+ — 45.4% mIoU	SAM — 10.2% mIoU

Recap: Capstone Stage 1

Stage 1: Key Conclusions

Foundation models showed better cross-domain robustness than task-specific models

- Their large-scale diverse training enables better generalization under real-world variations and unseen environments
- Using separate models for classification, detection, and segmentation increases deployment complexity
- Integrating outputs from different task models is difficult and inefficient

Need for a unified solution: A single model that performs all three tasks reduces operational complexity and provides consistent representations across tasks.

Florence 2 : Unified Representation for a Variety of Vision Tasks

Model Selection: Florence-2 a unified vision foundation model for:

Classification

Detection

Segmentation

Why Florence-2?

- Unified multi-task backbone with shared visual representations
- Single inference pipeline reduces deployment complexity
- Lower orchestration and maintenance overhead compared to multi-model systems
- Better consistency across vision tasks
- Lightweight and more practical for real-world deployment
- Supports few-shot adaptation with minimal labeled data

Key Motivation

- Replace separate task-specific pipelines with a single unified vision framework
- Improve scalability, robustness, and deployment efficiency under cross-domain conditions

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Florence 2 : Unified Representation for a Variety of Vision Tasks

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- Florence-2 uses a unified encoder–decoder architecture for multiple vision tasks within a single framework
- Different tasks are handled through prompt-based instructions, enabling flexible task adaptation without separate models
- Shared visual and textual embeddings allow consistent understanding across classification, detection, captioning, grounding, and segmentation tasks

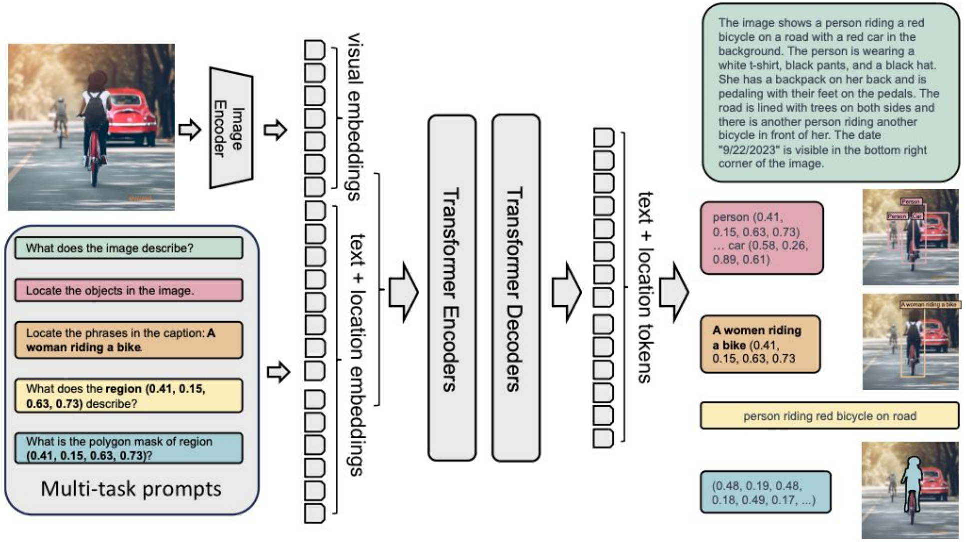

Figure 2. ***Florence-2*** consists of an image encoder and standard multi-modality encoder-decoder. We train ***Florence-2*** on our ***FLD-5B*** data in a unified multitask learning paradigm, resulting in a generaslist vision foundation model, which can perform various vision tasks.

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Florence 2 : Workflow Pipeline

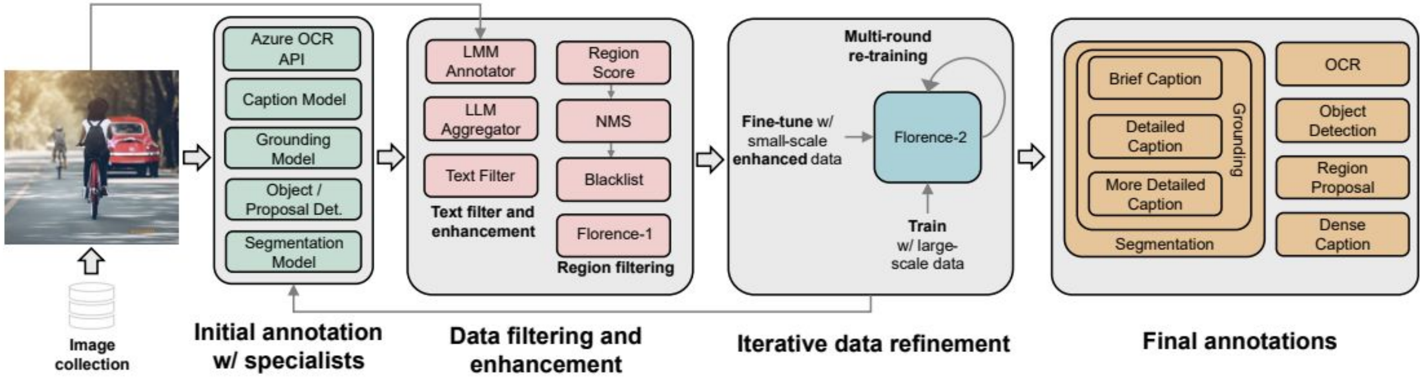

Figure 3. ***Florence-2* data engine** consists of three essential phrases: (1) initial annotation employing specialist models, (2) data filtering and enhancement to correct errors, and (3) an iterative process for data refinement. Our final dataset (***FLD-5B***) of over **5B** annotations contains **126M** images, **500M** text annotations, **1.3B** region-text annotations, and **3.6B** text-phrase-region annotations.

Initial Annotation: Specialist vision models generate captions, OCR, object proposals, grounding, and segmentation annotations

Data Filtering & Enhancement: LLM-based filtering, scoring, and refinement remove noisy or incorrect annotations

Iterative Data Refinement: Florence-2 is repeatedly trained and improved using large-scale refined data

Unified Multi-Task Learning: The final model learns shared representations across multiple vision tasks within a single framework

Final Outputs: Supports captioning, OCR, detection, grounding, region proposals, and segmentation using one unified model

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Experiment Matrix (Models × Tasks × Datasets)

Experiment Matrix : The unified Florence-2 + LoRA model is evaluated on all three tasks - classification, detection, and segmentation — across in-domain and cross-domain datasets. Stage 1 task-specific and foundation models are included as baselines to measure how much the unified model improves robustness and closes the lab-to-field gap.

Task	Specialist Models (Stage 1)	Foundation Model (Stage 1)	Unified Model (Stage 2)	In-Domain Dataset	Cross-Domain Dataset
Classification	ViT-Base, EfficientNet-B0, ResNet-50	CLIP	Florence-2 + LoRA	TrashNet (6 cls)	RealWaste
Detection	YOLOv8, Faster R-CNN	Grounding DINO	Florence-2 + LoRA	TACO (60 cls)	Trash-ICRA19
Segmentation	DeepLabV3+, U-Net, Mask R-CNN	SAM	Florence-2 + LoRA	TACO (polygons)	DWSD

Classification Results

What this shows: Accuracy and F1 of all classification models on TrashNet (in-domain, curated) and RealWaste (cross-domain, field-like). Compares task-specific CNNs/ViT, the CLIP foundation model, and unified Florence-2.

Model	Type	TrashNet (In-Domain) Acc / F1	RealWaste (Cross-Domain) Acc / F1
ViT-Base	Task-Specific	**96.44% / 0.964**	39.98% / 0.330
EfficientNet-B0	Task-Specific	89.53% / 0.890	32.89% / 0.245
ResNet-50	Task-Specific	80.83% / 0.802	17.85% / 0.128
CLIP	Foundation	67.83% / 0.670	42.68% / 0.380
Florence-2 Unified	Multi-Task	85.24% / 0.7008	**56.68% / 0.4769**

Analysis: Task-specific models peak in-domain but collapse cross-domain (ViT drops ~56 pts). Florence-2 trades a little in-domain accuracy for the best cross-domain performance, beating even CLIP by +14 pts.

Object Detection Results

What this shows: Detection performance on TACO (in-domain, 60 waste classes) and ICRA19 (cross-domain, underwater debris). Compares task-specific detectors (YOLOv8, Faster R-CNN), the Grounding DINO foundation model, and unified Florence-2.

Model	Type	TACO (In-Domain)	ICRA19 (Cross-Domain)
YOLOv8	Task-Specific	**mAP@0.5 = 64.17%**	mAP@0.5 = 13.49%
Faster R-CNN	Task-Specific	mAP@0.5 = 19.25%	mAP@0.5 = 13.60%
Grounding DINO	Foundation	mAP@0.5 = 15.26%	mAP@0.5 = 42.70%
Florence-2 Unified	Multi-Task	F1 = 36.57%	**F1 = 50.49%**

Analysis: YOLOv8 dominates in-domain but loses ~50 pts cross-domain. Florence-2 stays stable across domains and tops every model on ICRA19 (F1 = 50.49%).

Segmentation Results

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What this shows: Pixel-level segmentation quality (mIoU) on TACO polygons (in-domain) and DWSD (cross-domain). Compares task-specific segmenters (DeepLabV3+, U-Net, Mask R-CNN), the SAM foundation model, and unified Florence-2.

Model	Type	TACO mIoU (In-Domain)	DWSD mIoU (Cross-Domain)
DeepLabV3+	Task-Specific	**0.4541**	0.0483
U-Net	Task-Specific	0.3293	0.0630
Mask R-CNN	Task-Specific	0.2885	0.0842
SAM	Foundation	0.0380	0.1023
Florence-2 Unified	Multi-Task	0.2223	**0.2214**

Analysis: Specialist segmenters collapse cross-domain (DeepLabV3+ drops ~90%). Florence-2 has minimal domain drop (0.2223 → 0.2214) and beats every baseline on DWSD.

Key Findings

- Task-specific models win in-domain but collapse cross-domain (50–90% drop).
- Foundation models generalize better but lag in-domain accuracy.
- Florence-2 Unified sweeps all 3 cross-domain benchmarks: +14 pts classification, +13 pts detection, +116% segmentation.
- Near-zero domain drop in segmentation (0.2223 → 0.2214) — strongest generalization observed.
- One model outperform 15 specialists with LoRA weights addition.

Practical Recommendations

- Deploy unified foundation models (Florence-2 + LoRA) for real-world waste systems where input conditions vary.
- Use task-specific models only when the deployment domain exactly matches training data (controlled sorting facilities).
- Prioritize cross-domain robustness over peak in-domain accuracy — field performance matters more than benchmark scores.
- Next steps: higher LoRA rank, more epochs, task-balanced sampling, and adding field data to training.

Playground: Captioning

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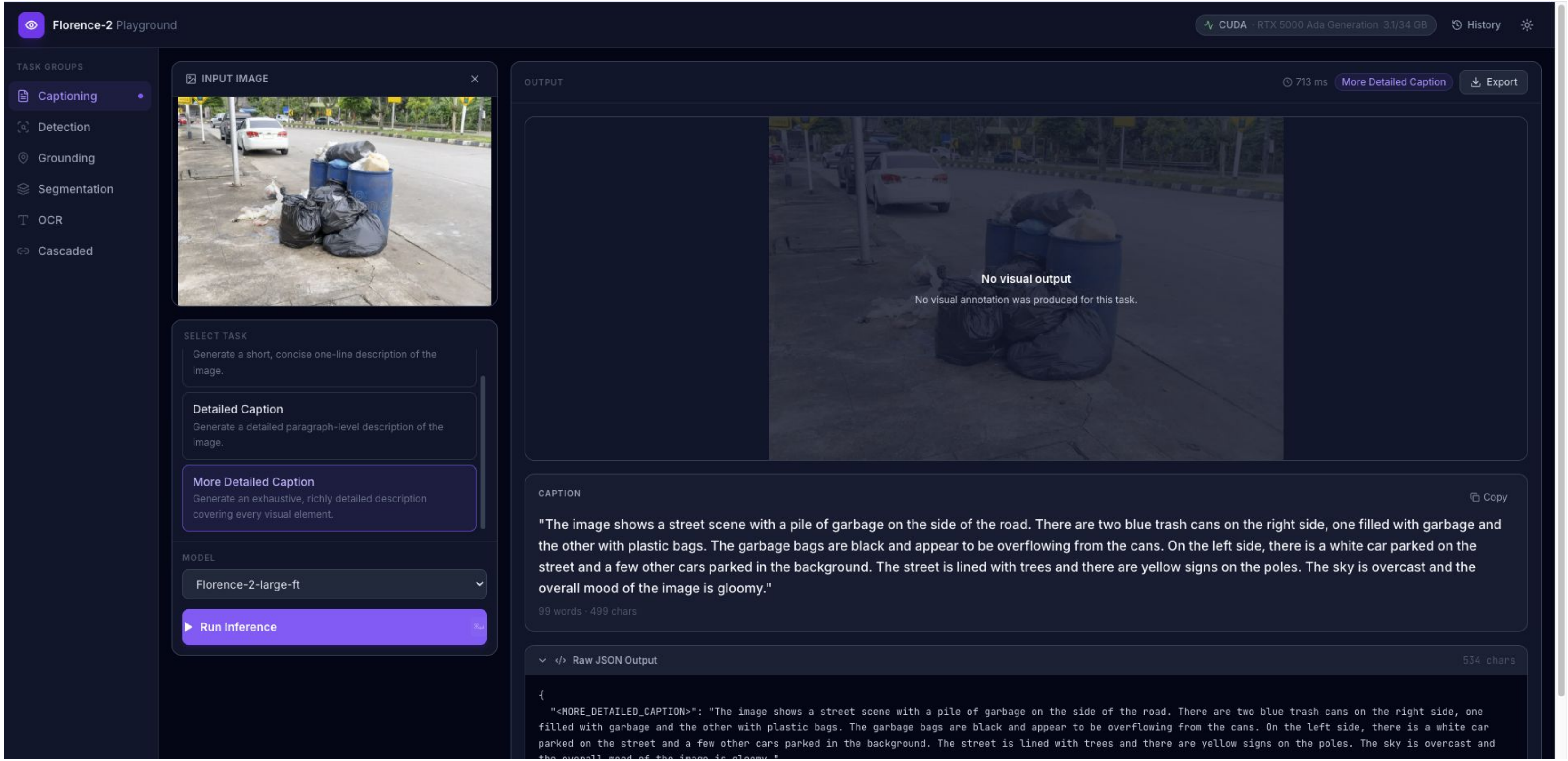
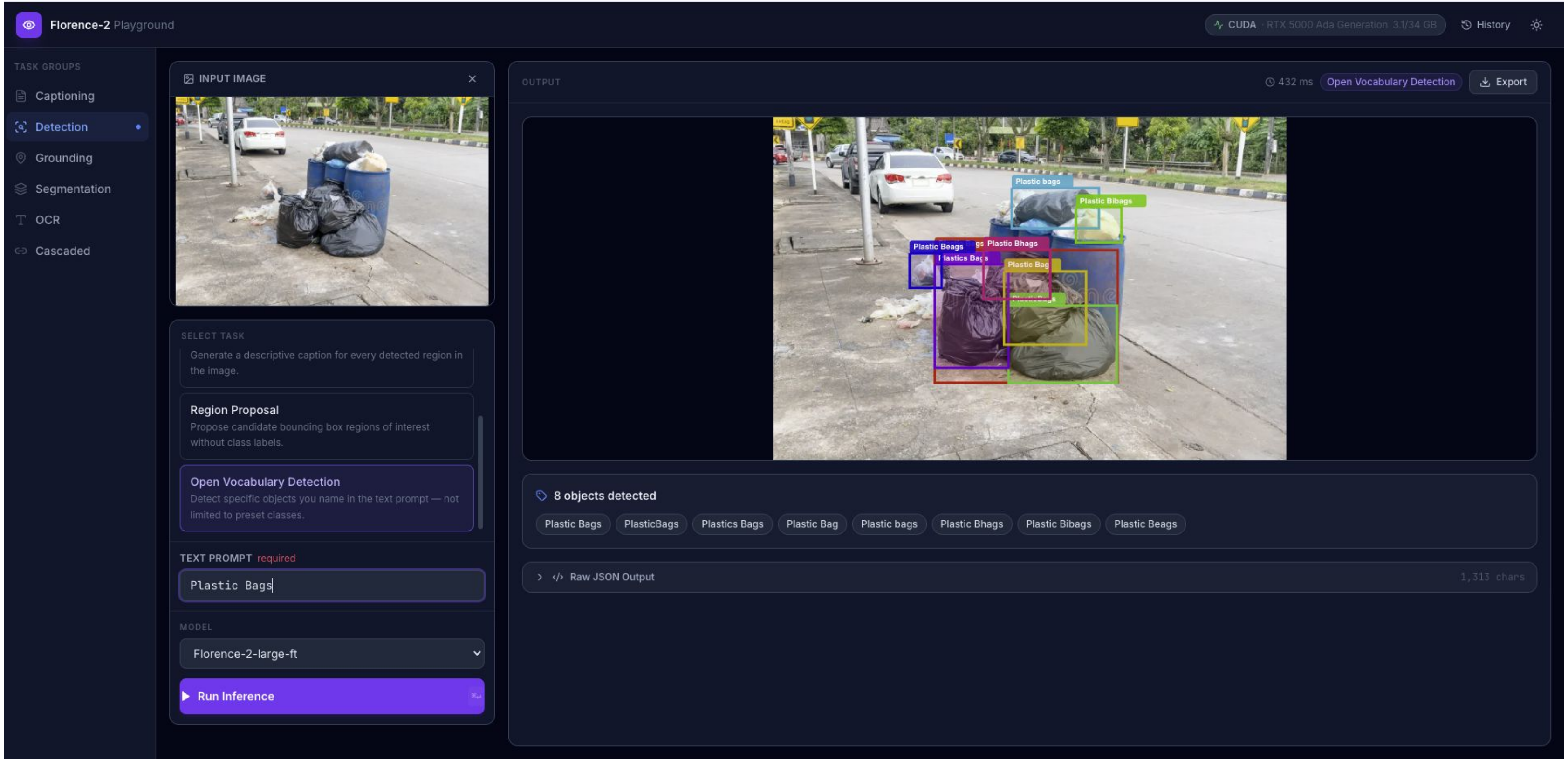
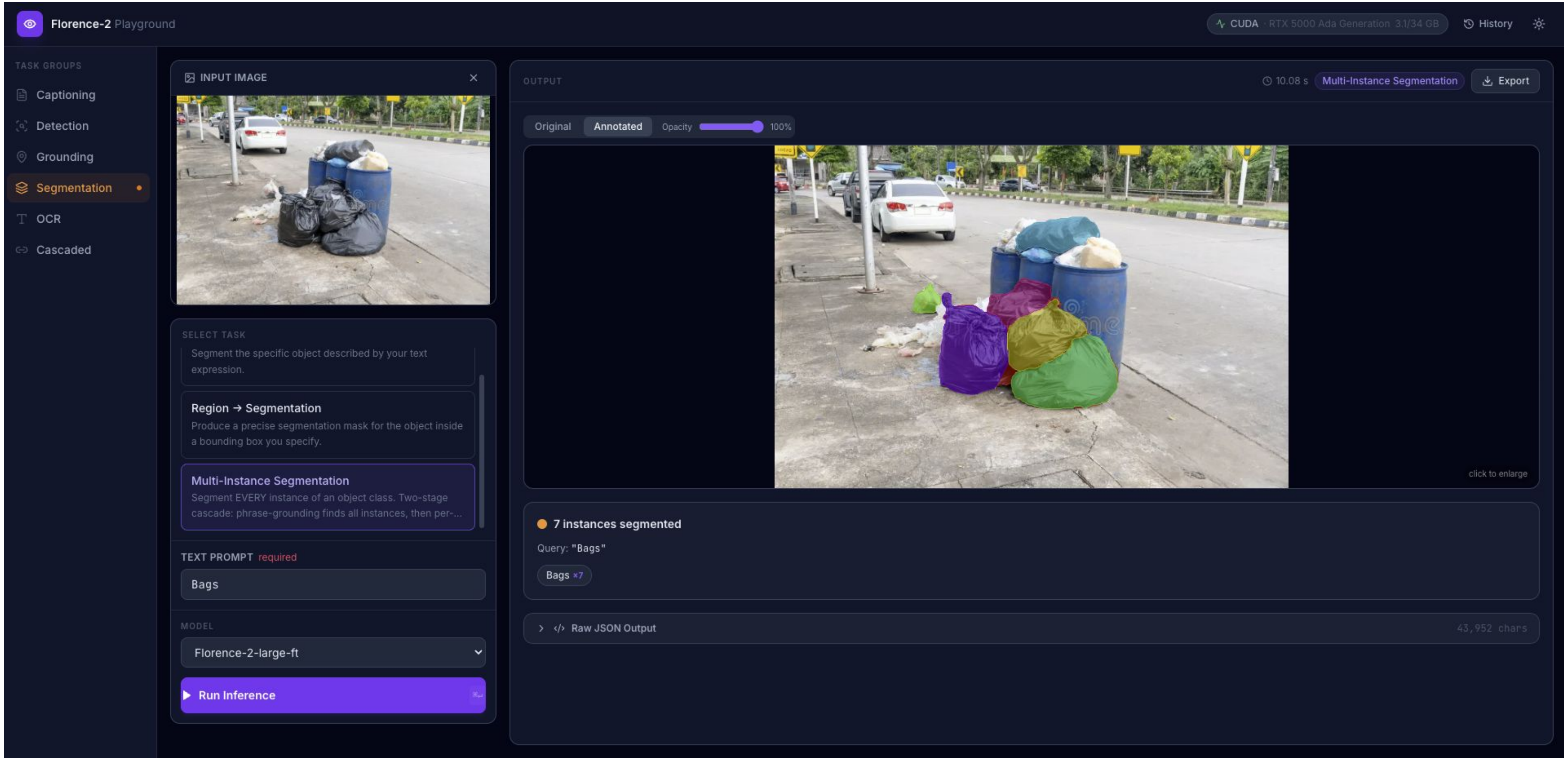


Playground: Object Detection



Playground: Segmentation



Conclusion

- Explored Florence-2 as a unified vision-language foundation model for waste management — one model, three tasks (classification, detection, segmentation).
- Bridged the lab-to-field gap: Florence-2 + LoRA sweeps all 3 cross-domain benchmarks (+14 pts classification, +13 pts detection, +116% segmentation).
- One model outperforms 15 specialists with just ~25 MB of LoRA weights — foundation models + lightweight fine-tuning are the right recipe for real-world, deployable waste systems.
- Foundation models + lightweight fine-tuning are the right recipe for real-world waste management, trading peak in-domain accuracy for robust cross-domain generalization.

Future Scope

- **Add more datasets:** broaden coverage across waste streams, geographies, and capture conditions to strengthen generalization.
- **Multi-Modal Foundation Models:** extend vision-based models by incorporating additional modalities (e.g., sensor data or metadata) to improve robustness and decision-making in waste management systems.
- **Scale up training:** more epochs (3 → 10+), higher LoRA rank (16 → 64), task-balanced oversampling.
- **Real-world pilot:** integrate the unified model into a live waste-sorting or surveillance pipeline and measure operational accuracy vs. lab metrics.

References

[1] Bin Xiao, Haiping Wu, Weijian Xu, et al. Florence-2: Advancing a Unified Representation for a Variety of Vision Tasks, CVPR 2024.

[2] Edward J. Hu, Yelong Shen, Phillip Wallis, et al. LoRA: Low-Rank Adaptation of Large Language Models, ICLR 2022.

[3] Sourab Mangrulkar, Sylvain Gugger, Lysandre Debut, et al. PEFT: State-of-the-art Parameter-Efficient Fine-Tuning Methods, Hugging Face, 2022.

[4] Pedro F. Proença and Pedro Simões. TACO: Trash Annotations in Context for Litter Detection, arXiv:2003.06975, 2020.

[5] Mindy Yang and Gary Thung. Classification of Trash for Recyclability Status, Stanford CS229 Project, 2016. (TrashNet)

[6] Sam Single, Saeid Iranmanesh, and Raj Raval. RealWaste: A Novel Real-Life Data Set for Landfill Waste Classification Using Deep Learning, Information, 2023.

[7] Michael Fulton, Jungseok Hong, Md Jahidul Islam, and Junaed Sattar. Robotic Detection of Marine Litter Using Deep Visual Detection Models, ICRA 2019. (Trash-ICRA19)

[8] Aloïs Babé, Rémi Cuingnet, Mihaela Scuturici, and Serge Miguet. Generalization Abilities of Foundation Models in Waste Classification, 2023.

[9] Alec Radford, Jong Wook Kim, Chris Hallacy, et al. Learning Transferable Visual Models From Natural Language Supervision, ICML 2021. (CLIP)

[10] Shilong Liu, Zhaoyang Zeng, Tianhe Ren, et al. Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection, ECCV 2024.

Thank You